Customer Shopping Behaviour Analysis

1. Project Overview

This project analyses customer purchasing behaviour using transactional data from 3,900 transactions across multiple product categories. The objective is to identify patterns in customer spending, segmentation, product preferences, and subscription behaviour to support data-driven business decisions.

2. Dataset Summary

- **Total Records:** 3,900
- **Total Features:** 18

Key Data Points:

- **Customer Attributes:** Age, Gender, Location, Subscription Status
- **Transaction Details:** Item Purchased, Category, Purchase Amount, Season, Size, Colour
- **Behavioural Metrics:** Discount Applied, Promo Code Usage, Purchase History, Purchase Frequency, Review Rating, Shipping Type

Data Quality Issue:

- 37 missing values identified in the *Review Rating* column

3. Data Preparation & Exploration (Python)

Data cleaning and preprocessing were performed using Python to ensure reliability and consistency:

- **Data Ingestion:** Loaded dataset using pandas
- **Initial Analysis:** Evaluated structure and summary statistics using `.info()` and `.describe()`
- **Missing Value Treatment:** Imputed missing *Review Rating* values using category-wise median
- **Column Standardization:** Renamed columns to snake_case for consistency
- **Feature Engineering:**
    - Created **age_group** by binning customer ages
    - Derived **purchase_frequency_days** to quantify buying behaviour

- **Data Validation:** Identified redundancy between `discount_applied` and `promo_code_used`; removed unnecessary column
- **Database Integration:** Exported cleaned dataset to MySQL for further analysis

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

4. Data Analysis (SQL – Business Insights)

Structured queries were used to extract key insights:

1. **Revenue by Gender:** Compared total spending across male and female customers

gender	sum(purchase_amou...
Male	157890
Female	75191

2. **High-Value Discount Users:** Identified customers using discounts but spending above average.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69

3. **Top-Rated Products:** Ranked products based on average review ratings

item_purchas...	average_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. **Shipping Impact:** Compared spending patterns between Standard and Express shipping.

shipping_type	round(avg(purchase_amount...
Express	60.48
Standard	58.46

5. **Subscription Analysis:** Evaluated revenue and average spend for subscribers vs non-subscribers

subscription_status	total_customer	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount Dependency:** Identified products heavily reliant on discounts for sales.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation:** Classified users into New, Returning, and Loyal segments

customer_segm...	no_of_custome...
loyal	3116
returning	701
new	83

8. **Top Products by Category:** Identified best-performing products within each category

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
2	Accessories	Sunglasses	161
3	Accessories	Scarf	157
1	Clothing	Pants	171
1	Clothing	Blouse	171
2	Clothing	Shirt	169
3	Clothing	Dress	166
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Purchase Behaviour:** Analysed correlation between frequent purchases and subscription likelihood.

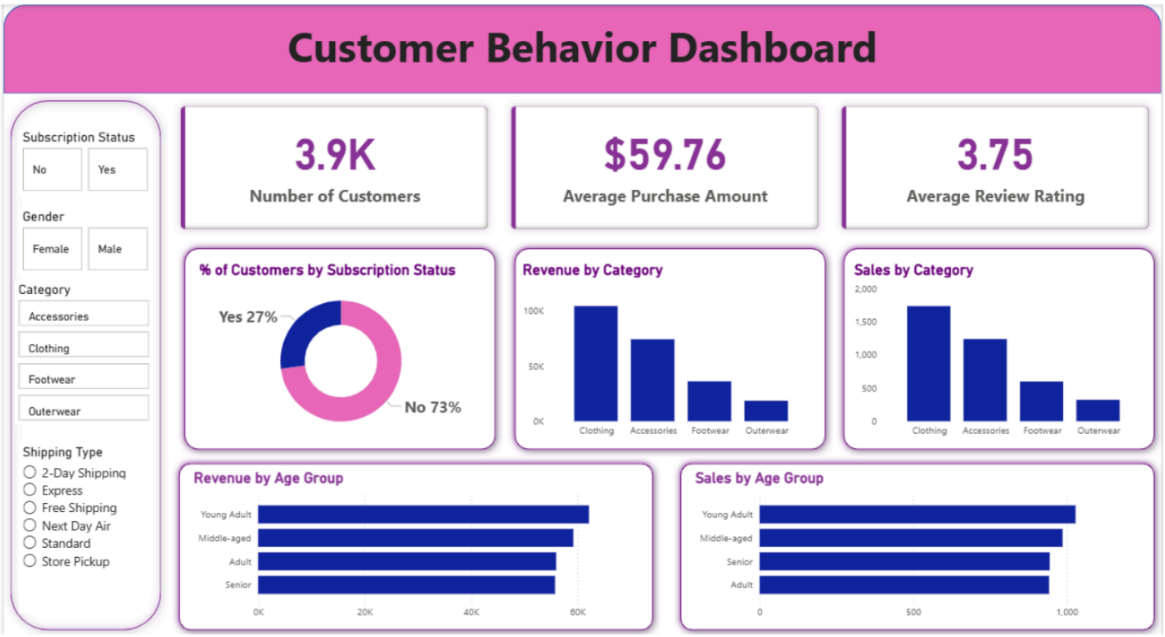
subscription_status	repeat_buyers
Yes	958
No	2518

10. **Revenue by Age Group:** Assessed contribution of different age segments to total revenue

age_group	total_revenue
young adult	62143
middle-aged	59197
adult	55978
senior	55763

5. Data Visualization (Power BI)

An interactive dashboard was developed to visualize key metrics, trends, and customer behaviour patterns, enabling intuitive and data-driven decision-making.



6. Business Recommendations

- **Increase Subscription Adoption:** Offer exclusive incentives to convert regular customers into subscribers
- **Strengthen Loyalty Programs:** Reward repeat customers to improve retention and lifetime value
- **Optimize Discount Strategy:** Balance promotional impact with profitability
- **Enhance Product Visibility:** Promote high-performing and top-rated products
- **Targeted Marketing:** Focus campaigns on high-value customer segments and express-shipping users